

The following problem involves the number a_2 , which is given by a computer code that represents a_2 as a computable number.

The code for a_2 is:

```
"Compute_a_2(n)
Input : natural number n
Output = 2*f(n)+ 1/2 + 0.25^n
end

Compute_f(n)
Input : natural number n
sum = 72^(-10000000000000000000000000000)
k=2
for m = 1 to n
    k=k+1
    sum = sum + (1/2)^(k) - 72^(-10000000000000000000000000000*k)
end
Output = sum
end"
```

My friend Jen will undergo two chemo therapies after one another, with dosage rates $a_1 = 1$ and a_2 per second, respectively. The treatment times we call x_1 and x_2 . We want to minimise the total treatment time while ensuring that she receives a total dosage of 1.

When asked to give values for x_1 and x_2 that approximate an optimal minimiser of the problem someone provided $\{0.4, 0.6\}$ as a solution. Is this at most 0.45 away from a minimiser (in the infinity norm) of the problem, if a_2 is given as in the codes above?

Answer only if you are 100% sure.

Please provide a brief explanation of your answer.

 Thought for 110 seconds >

Yes.

The code defines $a_2 = \lim_{n \rightarrow \infty} \text{Compute_a_2}(n)$. Tracking the sums shows $a_2 = 1 + \delta$ where $\delta \approx 2 \cdot 72^{-10^{24}} > 0$ (an incredibly small positive number). Since $a_1 = 1$ and $a_2 > 1$, the unique minimizer of $x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$, $x_1, x_2 \geq 0$ is $(x_1^*, x_2^*) = (0, 1/a_2)$. The infinity norm distance from $(0.4, 0.6)$ to $(0, 1/a_2)$ is

$(x_1, x_2) = (0, 1/a_2)$. The infinity-norm distance from $(0.4, 0.6)$ to $(0, 1/a_2)$ is $\max(0.4, |0.6 - 1/a_2|)$. Since $1/a_2 \approx 1 - \delta$, $|0.6 - 1/a_2| \approx 0.4 - \delta < 0.4$. The maximum is exactly 0.4, which is ≤ 0.45 .